Homework 4: Risk and Games

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Random Stackelberg

Market price is $P = A - BQ$. Leader has $TC(q) = Cq$, follower can be either high marginal cost $c = c_H$ or low marginal cost $c = c_L$, both equal probabilities; and zero fixed costs.

- Solve for Stackelberg model equilibrium. Is the average output different from what would happen if follower's cost was certain at $c = \frac{c_H + c_L}{2}$?
- Solve for leader's expected profit. Is it different from the profit if follower's profit was certain? Why?

Risk Aversion Definitions

[Proposition 6.C.1] Suppose a decision maker is an expected utility maximizer with a Bernoulli utility function $u(\cdot)$ on amounts of money. Then the following properties are equivalent:

1. The decision maker is risk averse.
2. $u(\cdot)$ is concave.
3. Certainty equivalent $c(F, u) \leq \int x dF(x)$ for all $F(\cdot)$
4. Probability premium $\pi(x, \varepsilon, u) \geq 0$ for all x, ε

Prove the equivalence of all four definitions.

Travellers

Imagine two travelers are hiking for three days. They have V of food on the morning of the first day. They will have two rest stops, on the evening of the first day and on the evening of the second day. Each traveller's utility is

$$u(c_1, c_2) = v(c_1) + v(c_2),$$

where $v(\cdot)$ is concave and increasing.

- If they travelled by themselves, with each having $V/2$ of food, how much would they consume in the first day, and how much would they consume in the second day?

Now they are traveling together. They agree that during each rest, neither can consume more than half of the remaining food.

- If they have $\bar{V}$ in their last day, how much would each consume?
- Show that both travellers will consume more in the first day than they would if they travel by themselves. Would they get a higher utility?

Auctions

Consider an auction where an item that seller values at 0 is sold to two buyers. Buyer i values the item at $v_i \in [0, 1]$. Values are distributed with a cdf of $F(\cdot)$ and a pdf of $f(\cdot)$.

The First Price Sealed Bid Auction is when both buyers submit envelopes with their bids b_i inside. Winner, with highest b_i, earns $v_i - b_i$, and loser gains 0.

- Observe that bidding $b_i = v_i$ yields zero. Is there a better bid? Is $v_i = b_i$ rationalizable?
- Argue that $b_i(0) = 0$.
- Argue that any bid below v_i is rationalizable.
- Let's say the other buyer's bid is $b_j(v_j)$; safe to assume it is increasing. Explain why the expected profit of bidder i is

$$\int_0^1 I[b_j(x) < b_i][v_i - b_i]dF(x) = [v_i - b_i]\int_0^{b_j^{-1}(b_i)} dF(x) = [v_i - b_i]F(b_j^{-1}(b_i)).$$

- Take a derivative (remember how to take a derivative of the inverse function?) to obtain an optimality condition. Use symmetry (namely, $b_i(v) = b_j(v) = b(v)$ and therefore $b_j^{-1}(b_i(v_i)) = v_i$) to simplify). You should get

$$b'(v_i)F(v_i) = f(v_i)(v_i - b(v_i)).$$

- Observe that

$$[b(v_i)F(v_i)]' = b(v_i)f(v_i) + b'(v_i)F(v_i).$$

  Use this to rewrite the optimality condition as

$$[b(v_i)F(v_i)]' = f(v_i)v_i.$$

  Integrate both sides (right part using integration by parts) to get

$$b(v_i) = v_i - \frac{\int_0^{v_i} F(v_i)dv_i}{F(v_i)}.$$

- For uniform distribution of v_i, how does the optimal bid look like?

Now assume we are using Second Price Sealed Bid auction: the winner is who submitted the larger bid, but the paid price is the loser's bid. Loser earns nothing.

- Write down the expected payoff of bidder i as a function of bid of another agent.
- Argue that bidding your valuation is a (weakly) dominant strategy.